

SIMCOE, ON, Canada

WMO: 715274

Lat: 42.850N

Lon: 80.270W

Elev: 789

StdP: 14.28

Time Zone: -5.00 (NAE)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	1.1	5.0	-8.1	3.7	2.6	-3.5	4.7	5.8	21.6	31.2	19.6	26.7	5.1	270	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	86.1	73.7	83.0	71.4	80.5	69.9	76.4	82.9	74.2	80.2	72.1	77.8	7.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.5	132.8	80.1	72.3	123.3	77.9	70.4	115.1	75.5	40.4	82.9	38.3	80.3	36.4	77.8	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	16.4	14.4	DB	-6.0	90.5	4.5	4.2	-9.3	93.5	-11.9	96.0	-14.5	98.4	-17.8	101.4	(4)
(5)			WB	-6.0	78.8	5.5	1.6	-9.9	80.0	-13.2	81.0	-16.3	81.9	-20.3	83.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.9	24.8	24.5	32.6	44.3	55.9	65.0	69.9	68.0	59.6	49.4	38.1	28.9	
	DBStd	18.22	10.53	9.88	10.65	8.31	8.03	6.81	5.60	5.64	7.35	7.79	8.36	9.64	
	HDD50	3430	781	713	551	212	29	1	0	0	9	110	369	655	
	HDD65	7046	1246	1133	1004	625	303	82	15	32	193	487	807	1119	
	CDD50	2289	0	0	12	40	212	450	617	559	296	90	12	1	
	CDD65	432	0	0	0	3	21	82	168	126	30	2	0	0	
	CDH74	3069	0	0	1	24	193	631	1285	810	120	5	0	0	
	CDH80	701	0	0	0	5	27	150	359	149	11	0	0	0	
Wind	WSAvg	5.8	7.6	6.6	7.0	6.7	5.6	4.8	4.0	3.5	4.5	5.4	7.0	6.3	
Precipitation	PrecAvg	38.7	2.7	2.3	3.4	3.6	3.0	3.4	3.0	3.1	3.7	2.9	3.3	3.9	
	PrecMax	46.8	5.5	5.2	6.7	5.5	5.4	6.3	6.1	10.7	9.3	6.8	6.5	6.5	
	PrecMin	29.1	0.6	0.4	0.6	1.9	1.3	1.3	1.3	1.2	0.4	0.6	1.3	1.4	
	PrecStd	4.2	1.3	1.3	1.7	1.2	1.1	1.3	1.3	2.1	2.2	1.5	1.4	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	51.9	71.3	77.7	83.2	88.2	92.5	87.3	81.2	74.3	64.0	55.3	
		MCWB	51.2	46.4	60.6	63.5	68.0	72.7	73.2	76.4	70.8	64.5	57.2	53.5	
	2%	DB	48.2	45.9	62.5	67.5	79.4	84.2	87.1	83.7	77.1	69.2	59.8	50.5	
		MCWB	46.5	41.4	53.6	56.8	66.3	70.5	74.8	73.7	68.7	60.8	55.6	47.9	
	5%	DB	41.9	40.4	54.5	62.5	75.3	80.6	83.7	81.0	74.4	65.7	56.1	44.4	
		MCWB	39.9	37.3	49.3	53.5	63.7	67.6	72.4	71.6	66.5	59.0	53.0	41.6	
	10%	DB	37.7	37.0	47.3	57.9	70.9	77.3	80.8	78.7	71.7	62.2	51.6	40.2	
		MCWB	35.9	35.1	42.5	50.0	61.4	66.6	70.9	70.2	65.1	56.8	48.4	37.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.6	47.8	61.2	63.9	71.5	77.0	79.7	78.8	73.6	66.2	59.7	55.0
			MCDB	53.4	49.7	70.2	75.5	78.0	85.0	84.1	83.6	78.7	71.7	62.1	55.1
2%		WB	46.7	41.3	54.8	59.4	68.9	72.9	76.8	76.2	70.6	63.3	56.6	47.8	
		MCDB	47.6	45.1	59.6	65.7	76.0	81.0	83.4	82.0	74.7	67.4	58.6	49.5	
5%		WB	40.4	37.9	49.2	54.9	65.5	70.5	74.6	74.0	68.8	60.6	53.2	42.1	
		MCDB	41.7	40.0	55.4	60.3	74.1	77.4	81.4	78.8	72.6	64.4	56.4	44.1	
10%		WB	36.1	35.1	42.9	50.8	62.0	68.3	72.4	71.9	66.8	57.7	48.6	38.1	
		MCDB	37.6	36.8	46.9	56.7	69.6	74.7	78.6	76.5	70.9	61.4	51.4	40.0	
Mean Daily Temperature Range		5% DB	MDBR	12.2	13.3	14.6	16.9	18.7	18.6	18.2	17.6	17.4	16.0	12.1	10.9
			MCDBR	14.7	15.8	24.3	23.5	24.2	22.1	20.7	19.0	18.4	19.5	16.8	14.7
	MCWBR		14.0	13.5	17.4	15.5	14.3	12.4	10.6	10.9	12.4	13.5	14.3	13.7	
	5% WB	MCDBR	14.7	15.1	23.5	21.2	22.1	17.9	18.8	17.2	17.0	16.7	17.0	14.7	
		MCWBR	14.6	13.6	17.7	15.8	13.7	12.8	11.5	11.4	13.3	13.2	15.5	14.3	
Clear-Sky Solar Irradiance	taub	0.293	0.343	0.334	0.373	0.394	0.431	0.424	0.409	0.377	0.357	0.311	0.291		
	taud	2.403	2.206	2.368	2.320	2.310	2.239	2.280	2.307	2.375	2.415	2.512	2.476		
	Ebn at Noon	269	267	288	283	279	268	269	269	270	261	261	260		
	Edh at Noon	24	35	34	39	40	43	41	39	34	29	22	21		
All-Sky Solar Radiation	RadAvg	460	676	1068	1425	1753	1941	1932	1695	1339	816	504	317		
	RadStd	40	83	128	152	169	130	135	88	124	77	72	36		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page